IRREGULARITIES – CHIEF (Coursework only)

If you suspect an assessment irregularity (e.g. plagiarism, collusion or multiple submission) please list the student ID and potential issue, otherwise leave blank.

STUDENT ID(s)	SUSPECTED ISSUE

NUMERICAL ANSWERS – CHIEF (Exam only, EXCEPT RESITS, published to students)

Question		Numerical Answer
1.	a)	Essay answer.
	b)	Syngas + Essay answer.
	c)	Essay answer.
	d)	$CH_\alpha O_\beta + w\ H_2O + m\ O_2 + x_6\ N_2$ (T=25°C) = $= x_1\ H_2 + x_2\ CO + x_3\ CO_2 + x_4\ H_2O_{(v)} + x_5\ CH_4 + x_6\ N_2$ + Essay answer.
	e)	Number of moles of water/mole biomass feedstock: 0.889 Number of moles of dioxygen/mole biomass feedstock: 0.354 Number of moles of output gas/mole biomass feedstock: 3.919
	f)	Molar fraction of H_2: 20.7% Molar fraction of CH_4: 0.2%
	g)	Molar feed rate of biomass: 1.04 mol/s Mass feed rate of biomass: 25.0 g/s
	h)	Molar LHV: 88.4 kJ/mol Volumetric LHV: 3.6 MJ/m^3 + Comment.
2.	a)	W_{RR} = 3.85 MJ
	b)	W_{Acc} = 5.51 MJ
	c)	W_{Aero} = 4.35 MJ
	d)	η_{TH} = 0.280
	e)	7.00 litres/100km

f)	Essay/proof answer. SFC, Mass, Aero, C_{RR} Importance of mass reduced by hybridisation, but ranking unchanged in this case.

ASSESSMENT CHECKLIST 2

MARKS FINALISATION – CHIEF AND PROOF

We confirm that:

- The marking and checking of the above assessment have been completed in accordance with the Manual of Assessment Procedures.
- The marks conform to the advertised marks matrix, Senate Marking Scale and appropriate UG/MScT Level descriptors.
- Marks have been entered onto the spreadsheet by the Chief, moderated by the Proof and checked by the Checker.

Chief: __David Sims-Williams_____**Date:** ____9/6/2020_____

Proof: ___ Sergii Veremieieiv _____ **Date:** __ 09/06/2020 ____

Assessment Average: 72.44%

Are marks within the expected bands for this level? Yes □ No X
(NOT required for resit examinations and examinations with less than five submissions)

Chief's Report (Exams and Coursework, published to students and external examiners – NOT required for resit examinations)

- Brief discussion (if necessary) as to why marks are outside expected range.

The average is slightly above the top target band (within ~1%). The conditions under which the exam was taken allowed candidates to produce answers with few faults, producing high grades even against rigorous marking expectations. For reference, the combined average of the two cohorts taking this exam would be within the target band.

Cohort feedback:

Q1:
The students found this question relatively easy. Common errors were: not stating all assumptions made for syngas composition calculations, confusing the definition of molar fractions of syngas components (which is the number of moles of each component per one mole of syngas and not per one mole of biomass) and for the gasifier performance not considering effect of biomass moisture content on lower heating value of output syngas.

Q2:
Candidates performed well on this question. Some candidates did not solve part c) correctly, either using the wrong velocity, or introducing some confusion between time and distance integration/averaging. Most candidates solved part d) (engine thermal efficiency) but many used a longer route than required, unnecessarily involving the energy consumption for the drive cycle.

Chair BoE: ___Claudio Balocco_____ **Date**___12/6/2020___